EduConnect: Web Application

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EduConnect: Web App

BS in Computer Science-2025

A project submitted in partial
fulfillment of the requirements for
the award of the degree of BS in
Computer Science

UNIVERSITY OF EDUCATION LAHORE
MULTAN CAMPUS

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It is notified for the nomination of all the concerned that Mr. Saad Naeem, Usman Ghani, and Akash Javed BS student of Information Sciences of University of Education has completed all the requirements for the award of BS Degree in the discipline of Computer Science as technology as per detail given here under:

Cumulative Result

BS in Computer Science **Credit Hours:_____**

Registration No	Complete Name	Course work	Project	Total	Cumulative Grade Point Average (CGPA)

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Controller of Examination

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ABSTRACT

EduConnect is a web-based Learning Management System (LMS) designed to enhance the educational experience by bridging the communication gap between students and teachers in an online environment. The platform allows teachers to create and manage courses, upload learning material, conduct quizzes, and interact with enrolled students through dedicated discussion boards. Students can browse available courses, enroll in them, access course content, and participate in interactive learning activities.

Background:

The rapid shift toward digital learning in recent years has exposed several limitations in traditional educational systems, including inflexible structures, geographical barriers, and limited access to quality resources. While several online learning platforms exist, many fall short in delivering an integrated solution that supports real-time interaction, streamlined course management, and secure payments. This growing demand for accessible, feature-rich, and scalable e-learning solutions forms the basis for this project.

Objectives:

The primary objective of **EduConnect** is to design and develop a robust online education platform using the MERN stack (MongoDB, Express.js, React.js, Node.js). The platform aims to:

i. Enable teachers to create, manage, and monetize online courses.
ii. Provide students with a seamless learning experience through interactive content and progress tracking.
iii. Incorporate secure payment integration and automatic certification.
iv. Facilitate real-time communication between students and instructors.
v. Ensure responsive and user-friendly design for broader accessibility.

Researches:

Through the development and testing of EduConnect, we observed that integrating modern web technologies allows for effective handling of dynamic educational content,

secure data operations, and scale able user interactions, significantly enhanced user engagement. Usability testing indicated high satisfaction among test users, with the system demonstrating strong performance in handling simultaneous operations such as messaging, video links.

Conclusion:

EduConnect successfully fulfills the intended objectives by offering a modular, scale able, and user-centric platform for online education. It bridges several gaps in existing systems by combining core educational functionalities with modern web development practices. The application serves as a prototype for next-generation e-learning systems, delivering a more personalized and interactive digital classroom experience.

Implications:

The platform has significant implications for educational institutions, private educators, and learners. It enables remote education without compromising interactivity, supports income generation for content creators, and can be extended to different educational models,higher education, and professional training. Additionally, **EduConnect** lays the foundation for integrating future innovations such as AI-powered learning paths and predictive performance analytics.

Limitations:

Despite its effectiveness**, EduConnect** has some limitations. It currently lacks native mobile app support, multi-language functionality, and AI-based adaptive learning features. Performance under high concurrent usage scenarios remains untested. Future work will focus on scalability enhancements, mobile compatibility, and incorporating intelligent recommendation systems.

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EduConnect Web Application

CHAPTER 1

Gathering & Analyzing Information

1. Introduction

EduConnect is a modern solution to these challenges. It is a full-stack web application aimed at transforming the digital learning experience for both students and teachers. Built using the powerful and popular MERN stack (MongoDB, Express.js, React.js, and Node.js), EduConnect combines course management, communication, certification, and payments—all under one roof. This chapter explores the concept behind EduConnect, the problems it addresses, and the technologies used to bring it to life.

In recent years, technology has dramatically reshaped the education landscape. With the shift from traditional classroom teaching to digital learning, especially after the global COVID-19 pandemic, online education platforms have become a cornerstone of modern learning. These platforms provide flexible, accessible, and cost-effective ways to educate students from all walks of life. However, many still face challenges such as lack of interaction, difficulty in Despite their advantages, many existing platforms still face several key challenges. Some of the most pressing issues include:

- **Difficulty in Content Management**: Teachers often struggle with organizing course materials, quizzes, and assignments in a seamless, user-friendly way.
- **Limited Customization**: Off-the-shelf solutions rarely provide the flexibility required to tailor the learning experience to specific academic or institutional needs.
- **Insecure Payment Gateways**: With the rise of online payments for courses and certifications, there's an increasing need for secure and transparent transaction systems.

EduConnect is designed to support a wide range of users, from independent educators and small coaching centers to large institutions looking to digitize their curriculum. Its modular architecture ensures scalability and customization based on varying educational needs.

This chapter outlines the vision and motivation behind EduConnect, provides an overview of the key challenges it addresses, and introduces the technologies used in its development. By integrating essential educational features into a cohesive platform, EduConnect strives to bridge the gap between traditional learning environments and the digital future of education.

1.1 EduConnect

EduConnect is an innovative online education platform designed to streamline and enhance the digital learning experience for both students and educators. It serves as a centralized hub where teachers can effortlessly manage courses, create engaging content, conduct assessments, and issue certifications, while students can access learning materials, interact with instructors, and track their progress in real time. With its user-friendly interface and robust feature set—including communication tools, secure payment integration, and progress analytics—EduConnect bridges the gap between traditional classroom learning and modern e-learning, making education more accessible, interactive, and efficient.

EduConnect is a user-friendly, full-featured web application developed to support online education. It empowers teachers to create and manage courses, conduct quizzes, communicate with students, and provide certifications. Students can join courses, interact with teachers, attempt quizzes, and monitor their progress—all from a single, integrated platform.

1.1.2 Features of EduConnect:

- Easy Course Creation and Management
- Live Chat for Real-Time Communication
- Auto-Graded Quizzes and Assessments
- Certification on Course Completion
- Secure Payment Gateway for Premium Courses
- Login and Role Management for Teachers and Students
- Clean and Responsive UI using React.**js**

1.2 Current landscape of E learning?

The e-learning industry is expanding rapidly with platforms like Udemy, Coursera, and edX leading the way. Despite their success, many users face limitations such as lack of interactivity, inflexible course formats, or poor user support. EduConnect was built to offer an interactive, secure, and flexible learning environment that can adapt to different user needs

1.3 Problem Statements

1.3.1 Challenges in Remote learning?

Many existing e-learning platforms do not offer effective collaboration, which affects student engagement and learning outcomes.

1.3.2 Communication Gaps

A major issue is the lack of instant communication between students and teachers. Students are often unable to get timely help, which disrupts learning.

1.3.3 Payment Security & Certification Issues

Without secure payment systems or verified certifications, users hesitate to enroll in paid courses. Trust is a major factor in online education.

1.4 Objectives

- Develop a centralized and user-friendly online education platform
- Facilitate course content upload and quiz management
- Implement real-time chat features for better interaction
- Provide certification and integrate secure payments
- Design a visually pleasant and intuitive user interface

1.5 Goals

- Deliver high-quality education through a responsive and interactive platform
- Simplify the learning and teaching process
- Ensure data privacy and secure transactions
- Bridge the communication gap between educators and learners
- Provide a seamless and aesthetically pleasing user experience

- **Enhance digital learning experience:**
  - The primary goal of EduConnect is to create a dynamic and engaging online education platform that supports both students and educators. By combining intuitive design with powerful features, EduConnect enhances the overall learning experience through accessible course materials, interactive tools, and seamless communication.
- **Simplify course and content management:**
  - EduConnect provides teachers with an easy-to-use interface to create, manage, and update courses. It eliminates the complexity of traditional content delivery by offering structured modules, multimedia support, and integrated assessments—all in one platform.
- **Bridge communication gaps:**
  - One of the key challenges in online education is the lack of interaction between students and teachers. EduConnect addresses this by integrating real-time communication tools such as messaging and discussion boards to foster better engagement and collaboration.
- **Support certification and progress tracking:**
  - **EduConnect** automates the process of issuing certificates upon course completion and provides detailed analytics to track student performance and progress. This not only motivates learners but also builds credibility for the courses offered.
- **Ensure secure and efficient payments:**
  - With built-in payment gateway integration, EduConnect allows institutions and independent educators to monetize their courses securely. This feature makes the platform a practical choice for educational entrepreneurship and scalable e-learning businesses.

1.6 Research Questions

- How can we build an engaging learning experience using the MERN stack?
- What functionalities improve student-teacher interactions?
- How can we ensure security and reliability in online education systems?
- How does visual design impact e-learning effectiveness?

- What are the key factors influencing the adoption of online education platforms in various educational settings?
- How does the use of artificial intelligence and machine learning in online education platforms improve the personalization of learning?
- What challenges do teachers face when transitioning to digital teaching methods, and how can platforms like EduConnect help?
- How do online education platforms impact student performance and learning outcomes compared to traditional classroom-based education?
- What are the ethical implications of using AI-driven grading systems and assessments in online education platforms?
- How can online education platforms be made more inclusive for students with disabilities or those requiring special accommodations?

1.7 MERN Stack Overview:

The MERN stack is a collection of JavaScript-based technologies used to develop powerful web applications:

- **MongoDB**: A document-oriented NoSQL database that stores data in JSON-like format.
- **Express.js**: A minimal and flexible Node.js web framework that provides robust features for building APIs.
- **React.js**: A JavaScript library used to build dynamic user interfaces efficiently.
- **Node.js**: A runtime environment that executes JavaScript code server-side.

1.8 Methodology

In the development of this system, the **Agile methodology** was adopted to ensure flexibility, continuous improvement, and active collaboration between developers and end-users throughout the project lifecycle. Agile promotes iterative development, allowing features to be designed, developed, and refined in cycles called sprints. After each sprint, feedback is gathered and incorporated into the next iteration, ensuring that the final product closely aligns with user requirements and expectations.

Some given as:

- Html5
- Css3
- JavaScript
- Tailwind.css
- React.js
- Node.js
- Exprss.js
- Mongo DB
- ✓ Figma

1.9 Reason for Chosen Methodology

We chose the Agile development approach because it breaks the project into smaller parts and allows regular updates. This means we can make improvements based on user feedback without starting from scratch. It's like building a product step-by-step with feedback at every step.

1.9.1 MongoDB

MongoDB is used for storing all the data in EduConnect, such as user details, courses, quizzes, and results. It's easy to scale and flexible, which helps us quickly adjust the structure if we want to add new features later.

1.9.2 Express & React?

Express makes it easier to manage server-side operations like APIs and handling user data. React helps build the front-end interface with smooth navigation and a dynamic look. Users can experience fast page updates and easy interactions.

1.10 Node JS

Node.js runs the backend server of EduConnect. It's fast and efficient, especially when handling real-time features like notifications and chat messages. It helps the app remain responsive even when many users are active at the same time.

CHAPTER 2

Software Requirement Specification

2 Software requirements specification:

2.1. Introduction

2.2. Purpose

The purpose of this SRS is to provide a clear and comprehensive outline of the requirements for the EduConnect project. This document is intended for academic evaluation and will guide the project team in designing and developing the application.

Additionally, the SRS acts as a reference point for the project's validation and verification processes. It ensures that all planned features align with the outlined requirements, enabling the team to identify and address any gaps or inconsistencies during development. Through this document, the team aims to present a well-structured approach to problem-solving, emphasizing their technical and organizational capability.

2. Scope

EduConnect is a web-based educational platform aimed at facilitating online learning. The platform provides tools for students to enroll in courses, track their progress, participate in assessments, and receive certificates. Teachers can manage course content, interact with students, and monitor their learning outcomes. The system ensures ease of use and accessibility while maintaining scalability for potential future growth.

EduConnect aims to demonstrate the team's ability to develop a real-world software solution that addresses modern educational challenges. It showcases an end-to-end approach to software development, from identifying user needs to delivering a functional and user-centric product. The project also emphasizes innovation by incorporating advanced features like automated certificate generation, secure payment gateways, and real-time communication, making it a standout example of effective problem-solving in the field of software engineering.

3. Definitions, Acronyms, and Abbreviations

- EduConnect: The e-learning platform under development.

- UC: Use Case.
- Functional requirements
- NFR: Non-Functional Requirement.
- CRUD: Create, Read, Update, Delete operations.

4. References

- IEEE 830 Standard for SRS.

5. Overview

This SRS document is organized into seven sections, providing a comprehensive framework for the development of EduConnect. It begins with an **Introduction**, outlining the project's purpose, scope, and key terminology. The **Overall Description** offers a high-level view of the platform, detailing its features, target users, constraints, and assumptions.

The **Specific Requirements** section specifies functional and nonfunctional requirements in detail, ensuring clarity and traceability. Visual aids, such as use case diagrams, data flow diagrams, and system architecture models, are included in the **Diagrams and Models** section to complement textual descriptions.

Finally, the document addresses performance constraints, validation methods, and appendices to ensure all aspects of the project are systematically defined and accessible to the development team and evaluators.

II. Overall Description

1. Product Perspective

EduConnect is an academic project using the MERN stack to deliver a modular, responsive, and user-friendly platform. It will operate independently, serving as a prototype for a scalable e-learning system.

2. Product Functions

Student Features

- Browse and search for courses based on categories, difficulty levels, or teacher ratings.
- Enroll in free or paid courses easily.
- Participate in quizzes to test knowledge and receive instant feedback.
- Track learning progress through dashboards that show lessons completed, quiz results, and certifications earned.
- Download certificates automatically generated after completing a course.

- **Teacher Features**

- Create, edit, and manage courses with multimedia content such as videos, PDFs, and presentations.

- Create interactive quizzes and assessments to evaluate student performance.
- View analytics to track student engagement and quiz results.
- Interact with students using discussion forums, answering their questions and providing feedback.

- **Admin Features**

- Manage user accounts, including approving or blocking users.
- Monitor courses, quizzes, and user activity across the platform.
- Moderate discussion forums to ensure a healthy and productive environment.
- Generate platform-wide usage reports to understand trends and system performance.

3. User Characteristics

Students: Most students will be beginners in using online learning platforms. They require a straightforward and user-friendly interface with clear navigation and easy access to courses resources.

- **Teachers**: Teachers using EduConnect will have moderate technical skills. The platform provides simple tools to help them create content, track student progress, and communicate effectively.
- **Admins**: Admins oversee the platform's operation. They may be academic supervisors or technical team members responsible for troubleshooting and maintaining system functionality.

4. Constraints

- **Project Timeline:** Development is limited to a specific timeframe, so only core features will be implemented in the first version.
- **Resources:** The team is working with minimal financial and technical resources.
- **Hosting:** The platform will initially use free-tier cloud hosting services, which may limit scalability and performance under high traffic.
- **Language Support:** Only English will be supported in this version, with plans for future multi-language integration.

5. Assumptions and Dependencies

- Tools and Technologies: Developers have access to essential tools like VS Code, Node.js, and MongoDB.
- Third-Party Libraries: The project assumes that third-party libraries (e.g., payment gateways, email services) will function reliably without major issues.
- Internet Stability: Both developers and users will require a stable internet connection for optimal performance.
- User Engagement: It is assumed that students and teachers will actively engage with the platform and provide feedback for improvements

Functional Requirements

No	Requirement	Description
FR-1	Course Enrollment System	**Student Side:** Browse available courses by category, subject, or teacher rating. Add courses to a wish list for future enrollment. Enroll in free or paid courses with payment confirmation for the latter. **Teacher Side:** Define course pricing or offer free courses. View enrollment statistics and course performance metrics.

FR3	Content Management System	Upload multimedia content (videos, and interactive presentations).
FR4	Discussion Forums	Allow thread creation under specific topics by teachers. Provide reaction options (like, dislike) for posts to encourage engagement.
FR5	Analytics and Reporting	**For Teachers:** Detailed analytics on course engagement, quiz performance, and student participation. **For Admins:** Platform-wide insights, including revenue, active users, and course popularity.

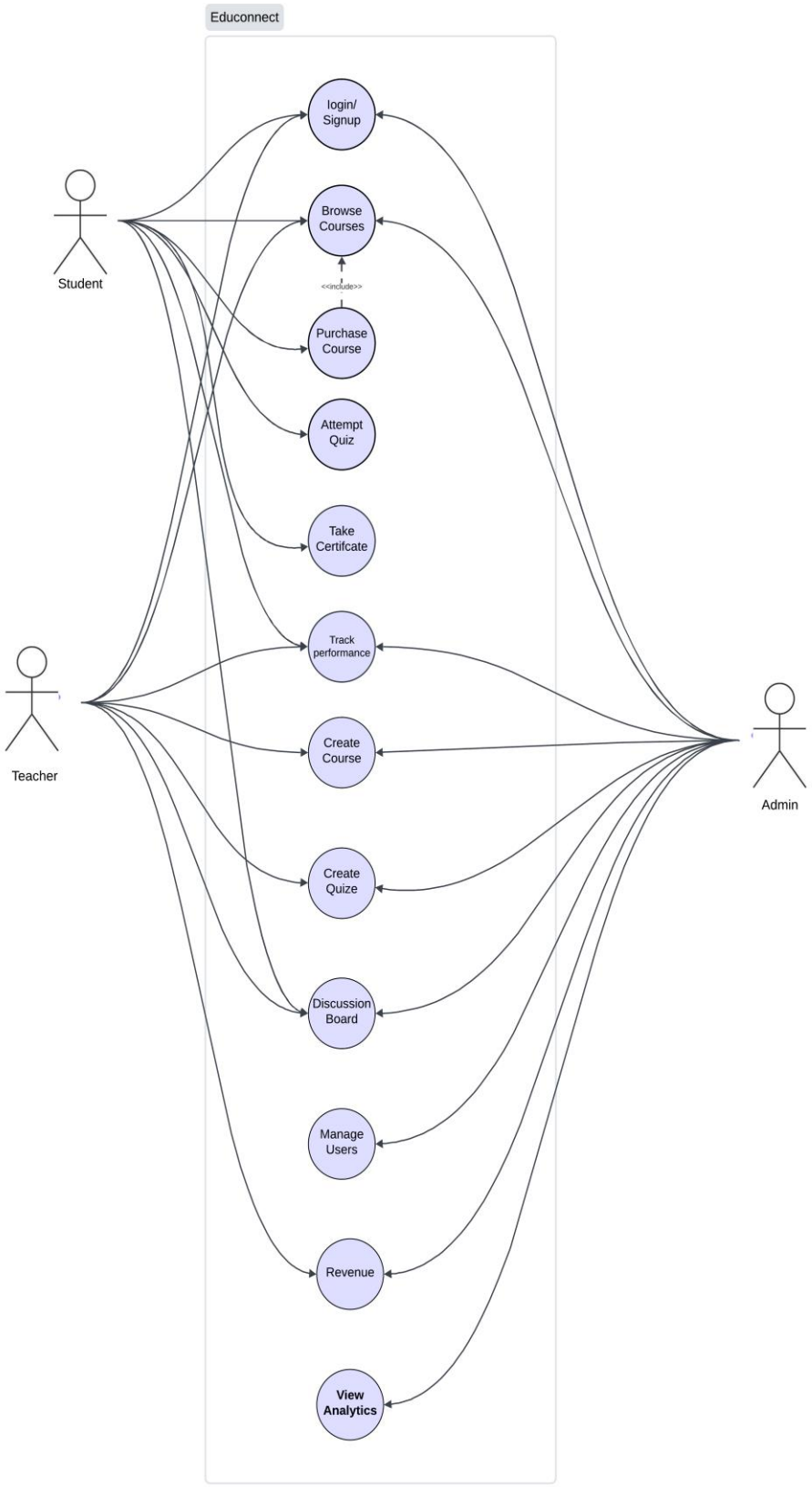
		For Students: Progress dashboards showing time spent on learning, grades, and certifications achieved.

Non-Functional Requirements

No	Requirement	Description
NFR-1	Scalability	Use cloud services like AWS or Azure to handle growth in traffic and data. Implement database sharing for efficient handling of large datasets.
NFR-2	Availability	Ensure 99.9% uptime with backup servers and disaster recovery protocols.
NFR-3	Data Privacy	Comply with data protection laws like GDPR for international users. Secure user data with encryption
NFR-4	Performance	Page load time under 3 seconds, even with high traffic. Real-time interactions (e.g., discussions) with minimal latency.
NFR-5	Maintainability	Modular codebase for easy updates and debugging. Version control using Git for collaborative development.
NFR-6	Cross-Browser Compatibility	Ensure the platform works seamlessly across popular browsers (Chrome, Firefox, Safari, Edge).
NFR-7	Accessibility	● Follow accessibility standards like WCAG 2.1 for users with disabilities. ● Include features like text-to-speech and adjustable font size

CHAPTER 3

Software Analysis



Use case diagram 3.1

Use Case Description

1. Register User

Attribute	Details
Use Case ID	UC01
Actors	Student, Teacher
Description	Allows students and teachers to create accounts on the platform to access relevant features.
Preconditions	The user must have valid personal details (e.g., email, name).
Main Flow	1. User navigates to the registration page. 2. Enters required details such as name, email, and password. 3. Submit the registration form. 4. System validates and saves the data. 5. System sends a confirmation email and displays a success message.

2. Login User

Attribute	Details
Use Case ID	UC02
Actors	Student, Teacher, Admin
Description	Enables users to log into the platform to access features based on their roles.
Preconditions	The user must have an existing registered account.
Main Flow	1. User navigates to the login page. 2. Enter email and password. 3. System authenticates the credentials. 4. System grants access and redirects to the user dashboard.

3. Create Course

Attribute	Details
Use Case ID	UC03
Actors	Teacher
Description	Allows teachers to create and upload new courses.
Preconditions	The teacher must be logged into the platform.
Main Flow	1. Teacher navigates to the course creation page. 2. Enters course details such as title, description, and pricing. 3. Uploads multimedia content (e.g., videos, PDFs). 4. Submit the form. 5. System saves the course and makes it available for students.

4. Enroll in Course

Attribute	Details
Use Case ID	UC04
Actors	Student
Description	Enables students to enroll in available courses.
Preconditions	The student must be logged in and have a valid payment method.

5. Discussion board

Attribute	Details
Use Case ID	UC05
Actors	Student, Teacher
Description	Allows students and teachers to engage in real-time discussions.
Preconditions	Both actors must be enrolled in the same course.
Main Flow	User navigates to the discussion board for the course. Posts a message or responds to an existing thread. System displays updates in real-time.

6. Take Quiz

Attribute	Details
Use Case ID	UC06
Actors	Student
Description	Enables students to complete quizzes for a course.
Preconditions	The student must be enrolled in the course.
Main Flow	1. Student selects a quiz from the course page. 2. Completes the quiz and submits responses. 3. System evaluates answers and provides instant feedback. 4. System records the results.

7. Take Certificates

Attribute	Details
Use Case ID	UC07
Actors	Student
Description	Issues certificates to students upon course completion.
Preconditions	The student must complete all course requirements.
Main Flow	1. System checks the student's progress and quiz results. 2. Verifies course completion status. 3. Generates and provides a downloadable certificate.

8. View Analytics

Attribute	Details
Use Case ID	UC08
Actors	Admin
Description	Allows admins to access platform-wide analytics and insights.
Preconditions	Data about user activity, course performance, and revenue must be available.
Main Flow	1. Admin navigates to the analytic section. 2. Selects metrics to view (e.g., course enrollment, quiz success rates). 3. Configures filters like time frame or user group. 4. Views and exports detailed reports.

CHAPTER 4
System Design

4.1 Introduction

System design is a critical phase in the development of EduConnect, laying the foundation for building a robust, scalable, and user-friendly online education platform. It involves planning how different components of the application—such as user interfaces, databases, servers, and APIs—will work together to meet the platform's goals.

EduConnect's system design focuses on creating an architecture that supports seamless interactions between teachers and students, ensures data security, and delivers real-time communication features. The design process translates user requirements and features into technical solutions, including database schemas, backend logic, and frontend components.

The architectural blueprint includes the use of the MERN stack (MongoDB, Express.js, React.js, Node.js), ensuring a modern, full-stack JavaScript solution that enables fast development, maintainability, and responsiveness. This design ensures that components like course creation, quizzes, chat, certifications, and payments are integrated efficiently.

By designing with flexibility and scalability in mind, EduConnect is prepared to support future enhancements and user growth while maintaining performance and reliability.

4.2 Purpose

EduConnect is an online learning platform designed to improve interaction between teachers and students by offering course-specific discussion boards, online quizzes, role-based access, and centralized academic content to support collaborative and efficient learning.

4.3 Scope

User Roles: The system supports two main user roles – **teachers** and **students**. Each role has specific access and features tailored to their needs.

Course Management: Teachers can **create and manage courses**, and students can **enroll** in them.

Review management: Teachers can provide feedback and reviews for student performance in their courses, while students can view and respond to these reviews through the platform.

Discussion Boards: Each course includes a **discussion board** where enrolled students and the respective teacher can communicate.

Access Control: Only authorized users (students or teachers) can access course content and discussion boards, ensuring **privacy and relevance**.

4.4 Architectural Representation (Architecture Diagrams)

To show the architecture of unique portal the DFD Data Flow Diagrams can be used.

4.4.1 DFD (Data Flow Diagram)

A data flow diagram is a visual tool for describing and examining how data moves through a system. These serve as the main resource and serve as the foundation for the creation of the other components. The logical transition of data from input to output through processing can be explained without reference to the system's physical components. The logical data flow diagrams are what these are called. The real tools and methods used to transfer data between individuals, offices, and workstations are depicted in the physical data flow diagrams. A collection of data flow diagrams serve as the complete description of a system. The data flow diagrams are created using two well-known notations, Yourdon, Gane, and Samson notation. Each element of a DFD is given a name that is illustrative. A number that will be used for identifying purposes is also utilized to better identify the process. DFD'S development takes place on a number of levels.

Each process in the lower level diagrams can be further delineated in the higher level DFD. The context diagram is another name for the lop-level diagram. It just has one process bit, but that process bit is crucial to understanding how the current system works. At the first level DFD, the process in the context level diagram is decomposed into various processes. Understanding at one level of information is exploded into greater

detail at the next level, according to the theory behind the process of expanding into more process. This is carried out up until the point at which further explosion is required and sufficient information has been provided for analysts to comprehend the procedure.

In order to describe system needs graphically, Larry Constantine originally created the DFD, which eventually led to the modular design. A DFD, commonly referred to as a "bubble chart," is used to define system requirements and indicate significant changes that will be turned into program in system design. As a result, it serves as the foundation for all design details. A DFD is made up of a number of bubbles connected by system data flows.

4.4.2 DFD Symbols:

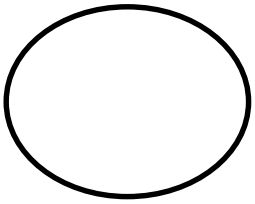
There are four symbols in the DFD.

A square designates a system's data source (originator) or destination.

Data flow is shown by an arrow. It is the conduit through which information is transmitted.

A process that converts an incoming data flow into an outgoing data flow is represented by a circle or a bubble.

A data store, data at rest, or transitory data repository is an open rectangle.



Process that transforms data flow

Source or Destination of data

Data Flow

Data Store

4.4.3 Constructing a DFD:

- Several rules of thumb are used in drawing DFD'S:
- Process should be named and numbered for an easy reference. Each name should be representative of the process.
- The direction of flow is from top to bottom and from left to right. Data traditionally flow from source to the destination although they may flow back to the source. One way to indicate this is to draw long flow line back to a source.
- An alternative way is to repeat the source symbol as a destination. Since it is used more than once in the DFD it is marked with a short diagonal.
- When a process is exploded into lower level details, they are numbered.
- The names of data stores and destinations are written in capital letters. Process and dataflow names have the first letter of each work capitalized

4.5 Data Flow

- A Data Flow has only one direction of flow between symbols. It may flow in both directions between a process and a data store to show a read before an update. The latter is usually indicated however by two separate arrows since these happen at different type.
- A join in DFD means that exactly the same data comes from any of two or more different processes data store or sink to a common location.
- A data flow cannot go directly back to the same process it leads. There must be at least one other process that handles the data flow produce some other data flow returns the original data into the beginning process.
- A data flow to a data store means update (delete or change).

DFD Diagrams Showing Architecture of the Project

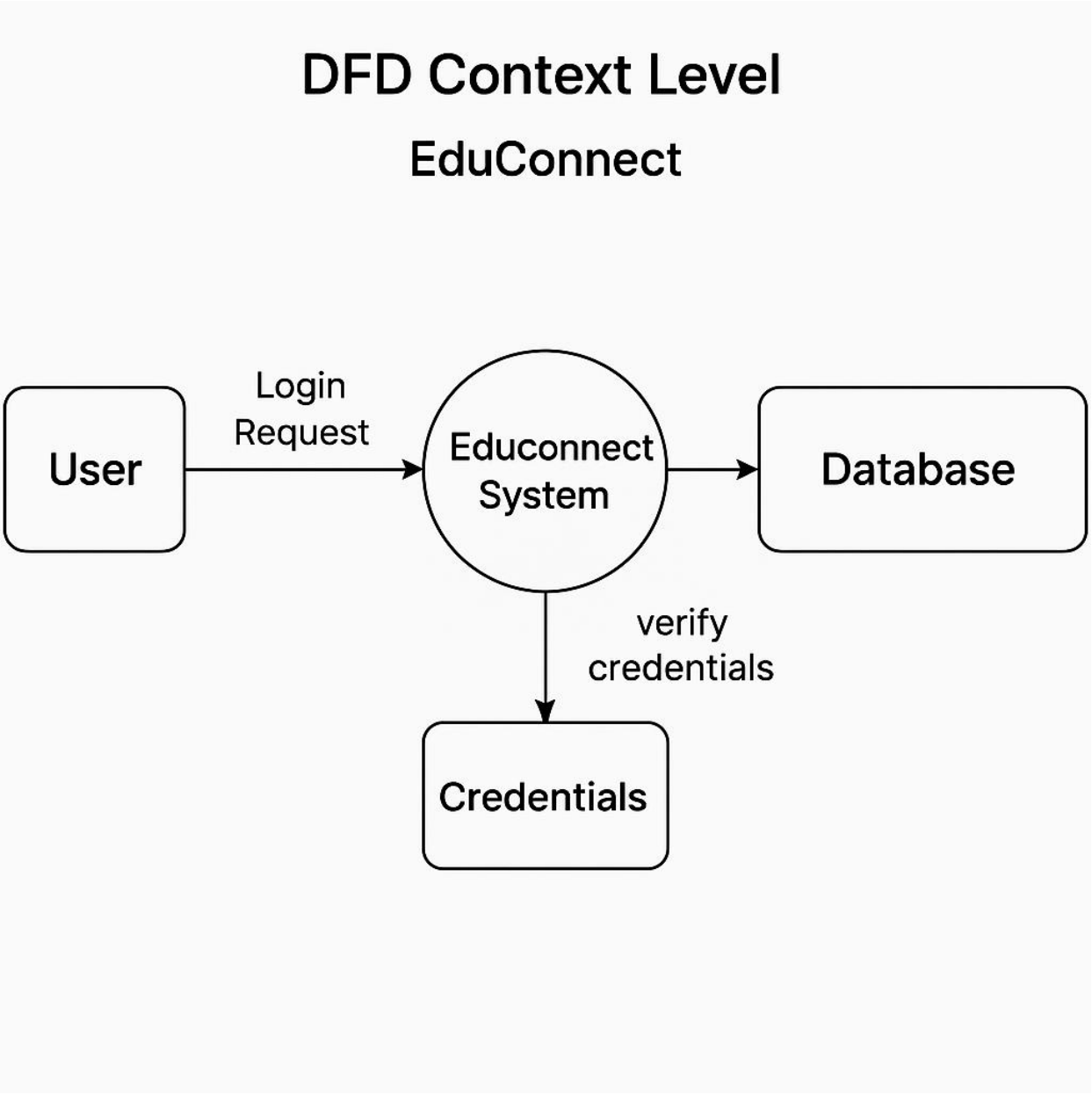


Figure 4.1 Level Context level

As shown in Figure 4.1, at the context level, the User interacts with the EduConnect system to log in, access features, and receive role-based services via the central system.

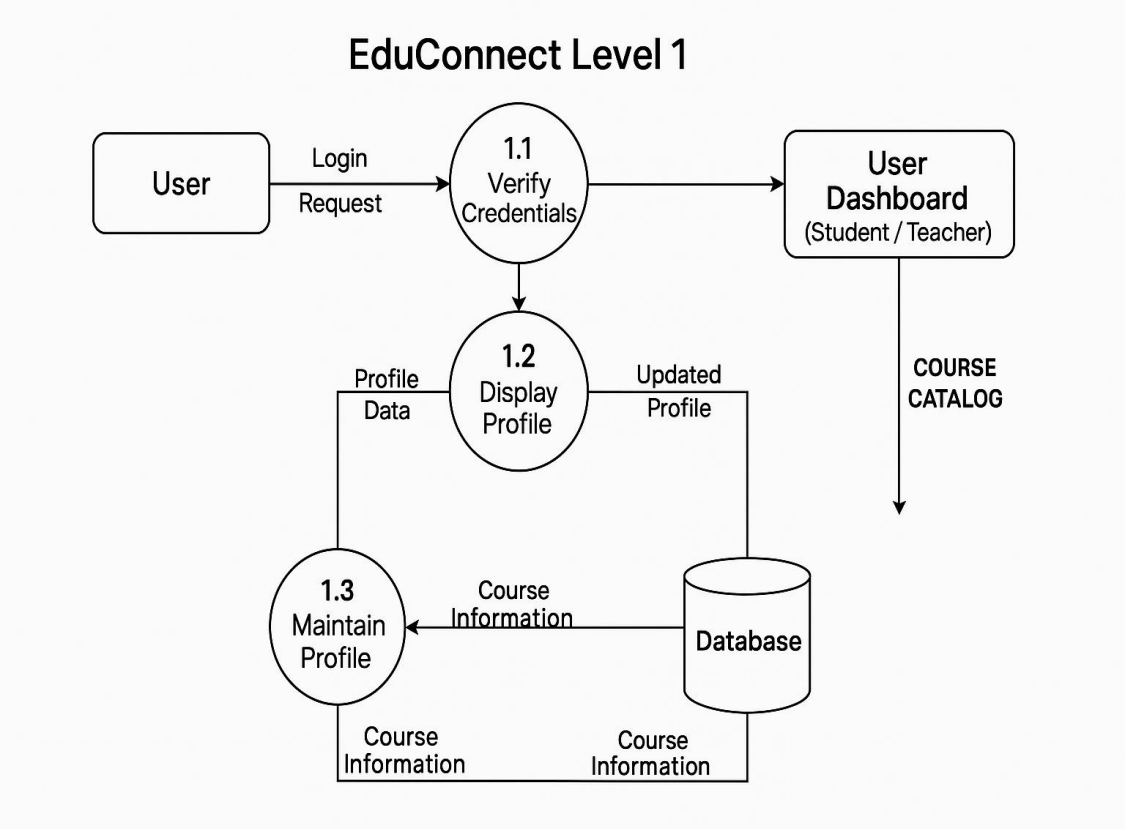


Figure 4.2

"As shown in Figure 4.2, at Level 1, the user interacts with core EduConnect modules such as Course Management, rating system, and Certification. The system processes user inputs, identifies the request, and responds with the appropriate output like course content, quiz results, or certificates."

4.5.3 Level 2

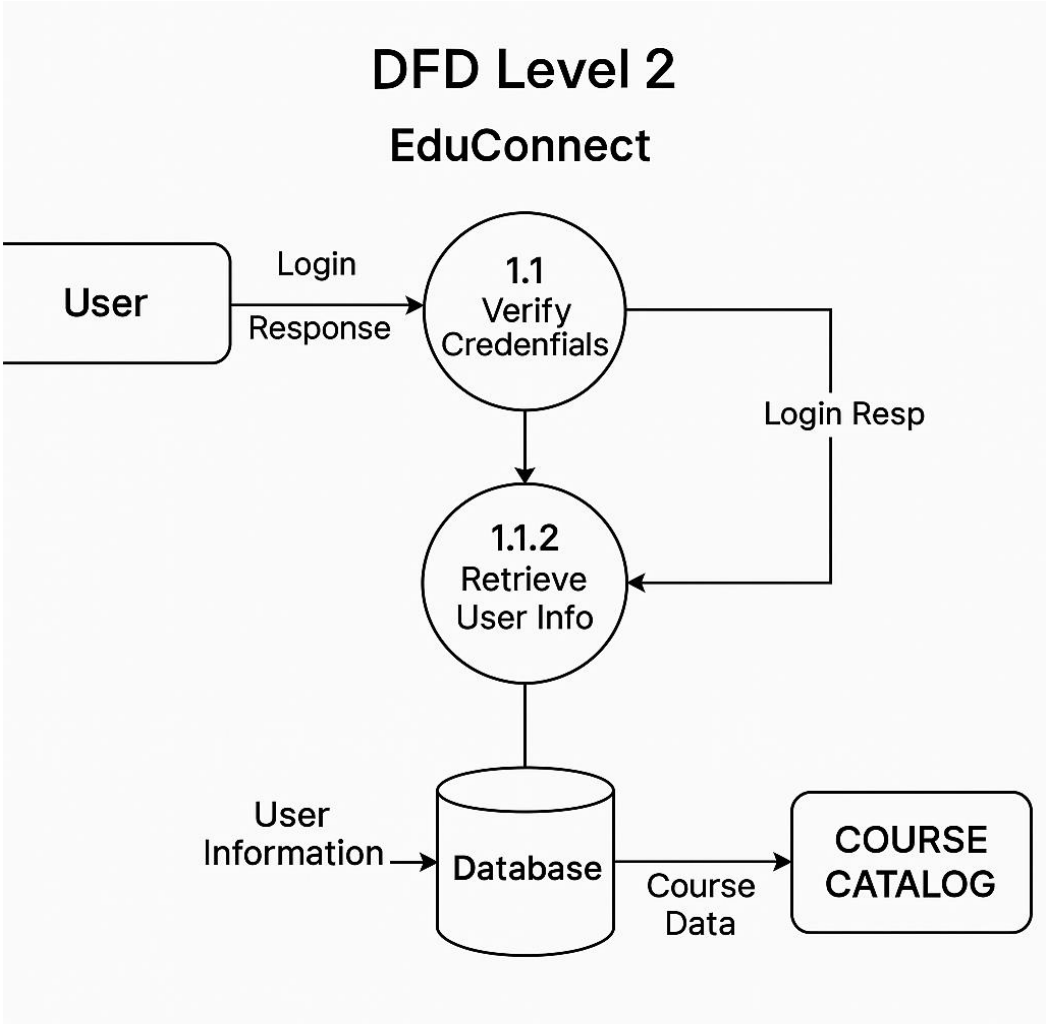


Figure 4.3 Level 2

As shown in Figure 4.3, the Level 2 DFD illustrates the *Verify Credentials* process, where the system checks login details against the database and grants access to the appropriate dashboard based on the user's role.

4.6 Class Diagram

An object model consists of the following important features:

- **Object Reference**
- In EduConnect, objects such as Student, Teacher, Administrator, and Course Management are accessed via object references. For example, a Teacher can invoke the manage Courses method through its reference, while a Student can call the take Course method.
- **Interfaces:** Interfaces are represented by method signatures such as upload Content(type: string) in the Course Management class, or register User(role: string) in the User Management class. These interfaces define what actions are possible, while the actual implementations are handled within the corresponding classes.
- **Actions:** Action in **EduConnect** is initiated when an object invokes a method in another object. For instance, when a Student object interacts with the Certification module, it may invoke award Certificate (). Similarly, the Administrator can monitor and oversee the platform by invoking methods like monitor Platform and oversee Users.
- **Exceptions:** Although not explicitly shown in the diagram, the system is expected to handle exceptions such as login failures, invalid quiz attempts, or failed payments gracefully using exception handling mechanisms.

✓ **Garbage collection:** In the JavaScript-based implementation of EduConnect, memory management is handled automatically. Objects like, Real Time Communication, or unused user sessions are cleared through garbage collection when no longer referenced, keeping the system efficient.

Here is Class diagram of EduConnect Web App

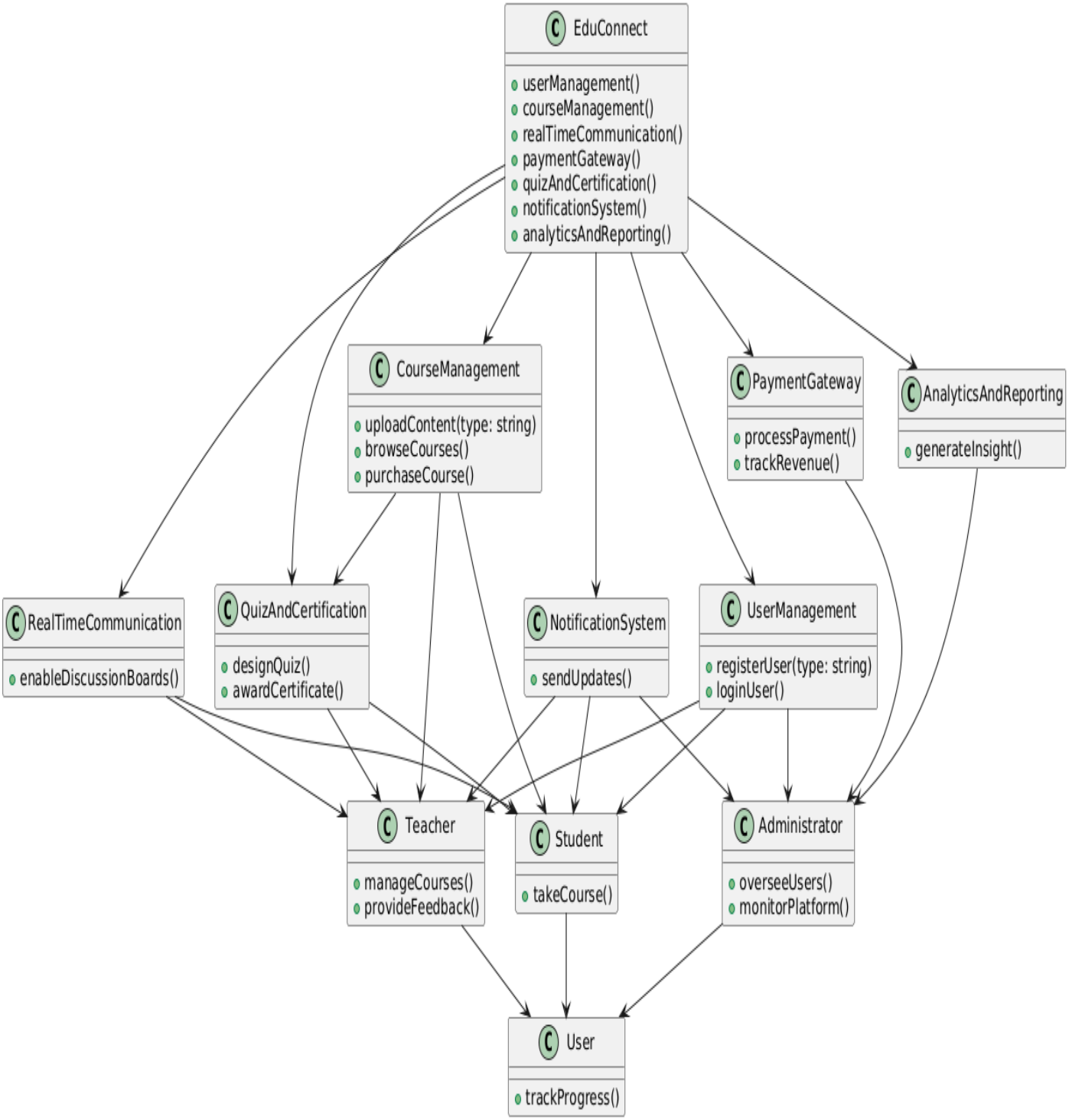


Figure 4.6 Class Diagram

EduConnect is built with modules like User Management, Course Management, Quizzes, and Real-Time Communication. Users register, access courses, attempt quizzes, and interact through discussion boards. Teachers upload content and design quizzes, while students take courses and receive certificates. Payments are managed securely, and analytics provide insights for platform monitoring. All components work together to deliver a smooth and interactive e-learning experience.

4.7 Sequence Diagram

A **sequence diagram** is a graphical representation that illustrates how different components of a system interact over time. In the context of **EduConnect**, the sequence diagram shows the flow of actions and communications between users (students, teachers, administrators) and system components (EduConnectSystem, CourseContent, PaymentGatewa).

At a high level, the sequence diagram for EduConnect includes the following components:

Users

- **Student**: Registers/logs in, browses and purchases courses, accesses course content, takes quizzes, posts queries, and provides feedback.
- **Teacher**: Registers/logs in, creates/uploads course content, manages courses, and responds to student queries.
- **Administrator**: Monitors platform activity, oversees transactions, generates reports, and flags content for review.

System Components

- **EduConnectSystem**: Acts as the core controller that handles user registration, login confirmation, content access coordination, and interactions across modules.
- **CourseContent**: Manages multimedia delivery such as videos, PDFs, and quizzes. It responds to course browsing and access requests.

- **PaymentGateway**: Handles the payment process when a course is purchased. It processes transactions and confirms payments back to the system.

Flow of Interactions

- All user types begin by registering and logging in through EduConnectSystem.
- Students browse courses, purchase one, and the payment is processed via PaymentGateway.
- After confirmation, EduConnectSystem enables course access and multimedia content is delivered from CourseContent.
- Teachers manage their courses by uploading content and responding to student queries.
- Students take quizzes and participate in discussions.
- Feedback and queries trigger real-time interactions between students and teachers.
- The administrator oversees activities, generates platform reports, and ensures content quality.

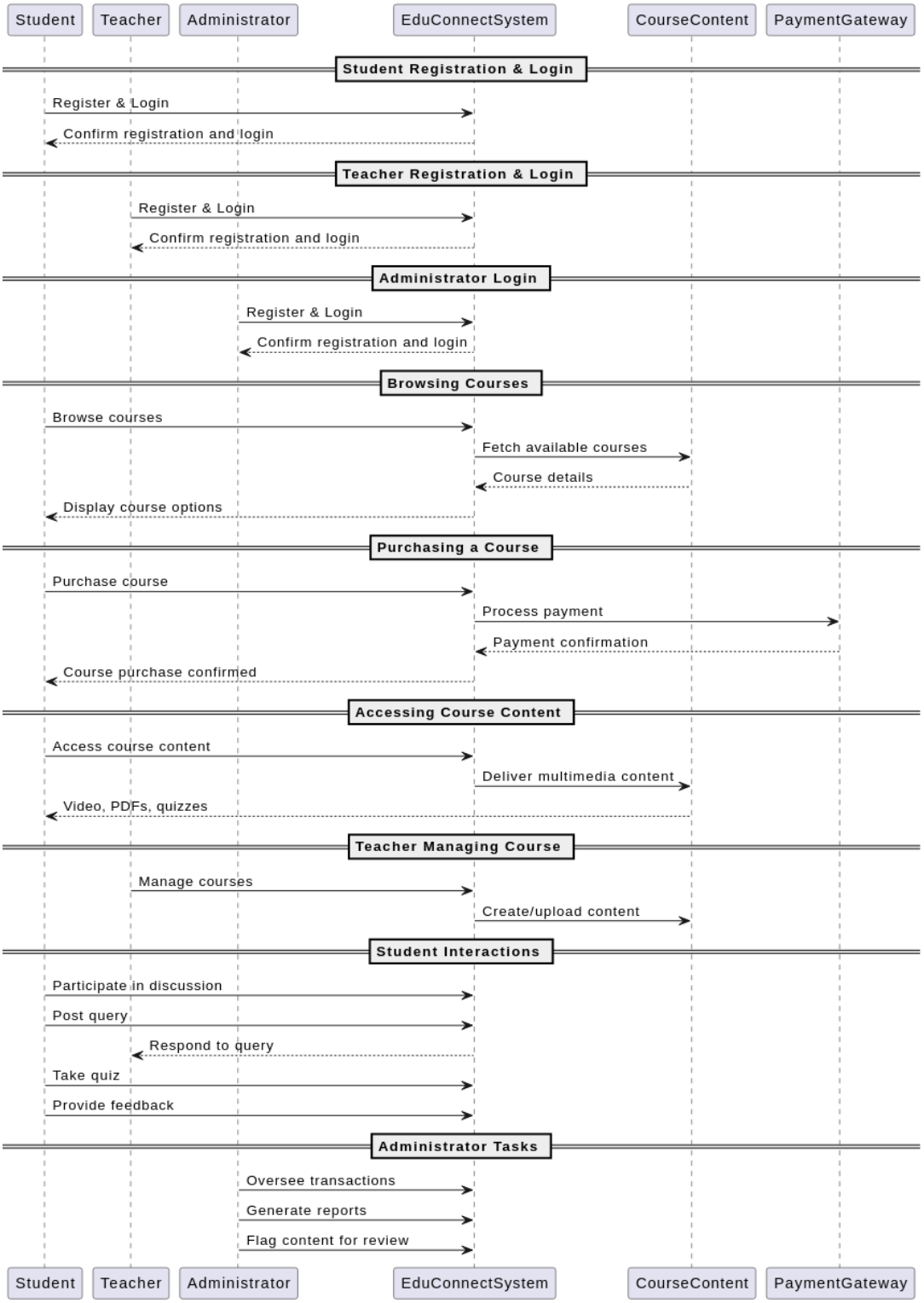


Figure 4.7 Sequence Diagram

4.10 Entity Relationship Diagram

An **Entity-Relationship Diagram (ERD)** is a data modeling technique that visually represents the key entities in the **EduConnect** system and how they are related. It serves as a **conceptual model** that lays out the data structure of the platform.

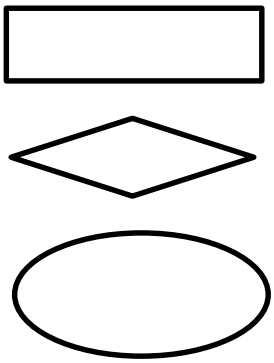
The components of an ERD include:

- ✓ **Entities**
- ✓ **Relationships**
- ✓ **Attributes**

Entity

Relationship

Attribute



Steps involved in creating an ERD include:

- **Recognizing and Classifying the Entities:**

The main entities identified in EduConnect include **User**, **Student**, **Teacher**, **Administrator**, **Course**, **Quiz**, **Discussion Board**, **Certificate**, and **Payment**.

- **Figuring Out How Each of the Entities Interact:**

For example, a **Teacher** creates a **Course**, a **Student** enrolls in a **Course**, Similarly, **Payments** are linked to **Courses**, and **Certificates** are awarded.

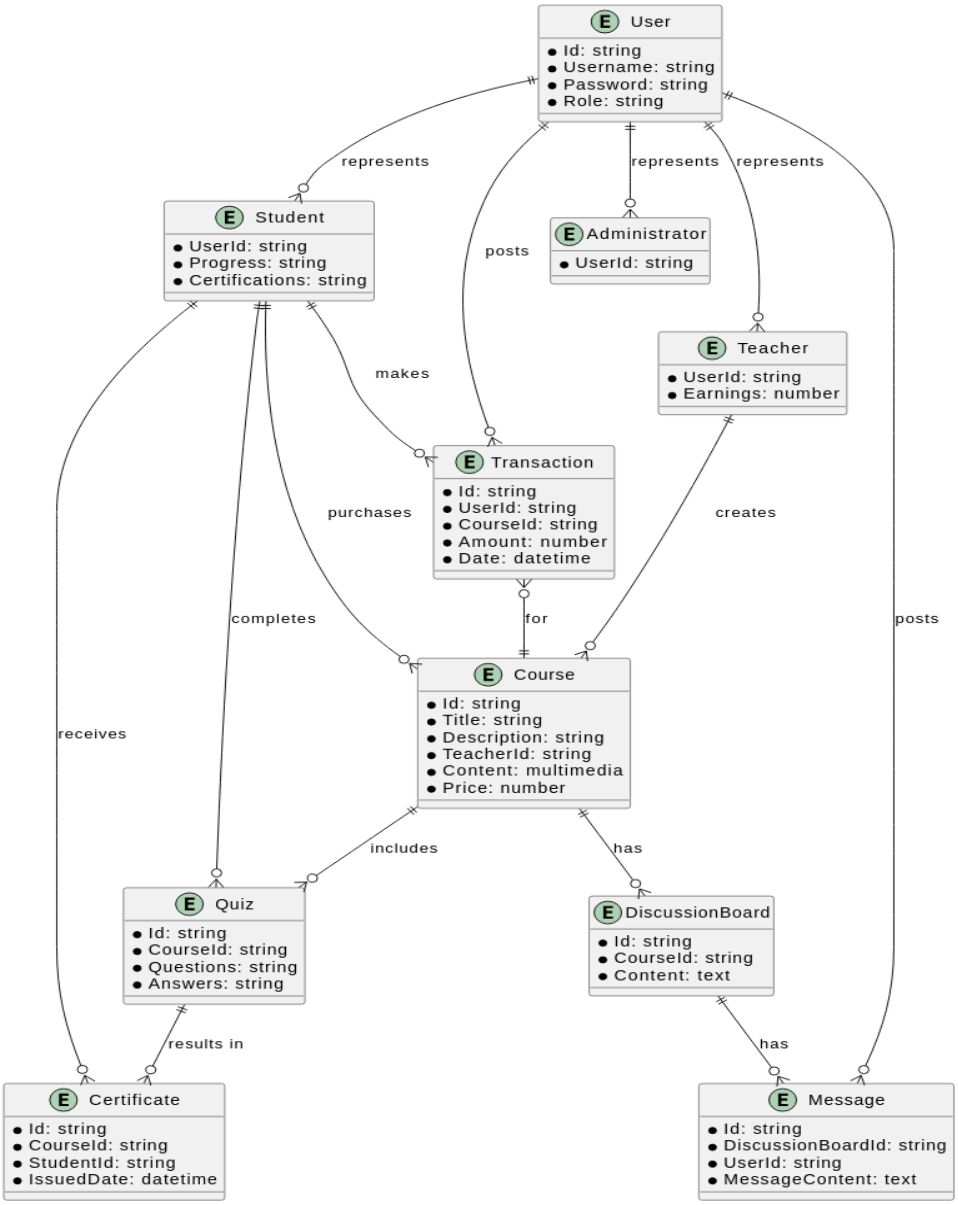
- **Analyzing the Interactions' Nature and Cardinality:**

A **Teacher** can manage multiple **Courses**, but a **Course** is managed by one **Teacher**. A **Student** can enroll in many **Courses**, and a **Course** can have many **Students**. These many-to-many and one-to-many relationships are captured in the ERD.

- **Designing the ERD:**

  Using the identified entities and their relationships, a visual ERD is created to show how data flows and is connected within EduConnect, ensuring a clear understanding of the system's data structure.

Figure 4.10 ERD



CHAPTER 5

Graphical User Interface

5.1 GUI (Graphical User Interface)

EduConnect features a clean, user-friendly GUI designed for students, teachers, and administrators. It offers easy navigation for course enrollment, content access, discussion participation, and platform management. The interface is responsive, visually interactive, and adapts across devices. Buttons, tabs, and menus provide real-time feedback, ensuring a smooth and accessible learning experience for all users.

5.1.1 What are the benefits of GUI?

The **GUI** in EduConnect makes the platform easy to use for all types of users—students, teachers, and administrators. It eliminates the need for technical knowledge, allowing users to navigate, access courses, manage content, and communicate effortlessly. The visually organized layout improves engagement, reduces errors, and enhances the overall learning experience through a modern, interactive interface.

5.1.2 How does the user interact with a GUI?

In EduConnect, users interact with the GUI using a mouse, keyboard, or touchscreen to perform actions like logging in, browsing courses, accessing learning materials, and participating in discussions. The interface includes clickable buttons, dropdown menus, tabs, and icons, allowing students, teachers, and admins to easily navigate the platform and complete tasks without any technical knowledge.

GUI (Graphical User Interface) of EduConnect using MERN Stack

This is the GUI of our project. It is developed by the

i. **Saad Naeem**

ii. **Usman Ghani**

iii. **Akash Javed**

This is web base application and run in MERN stack.

5.2 Home Page

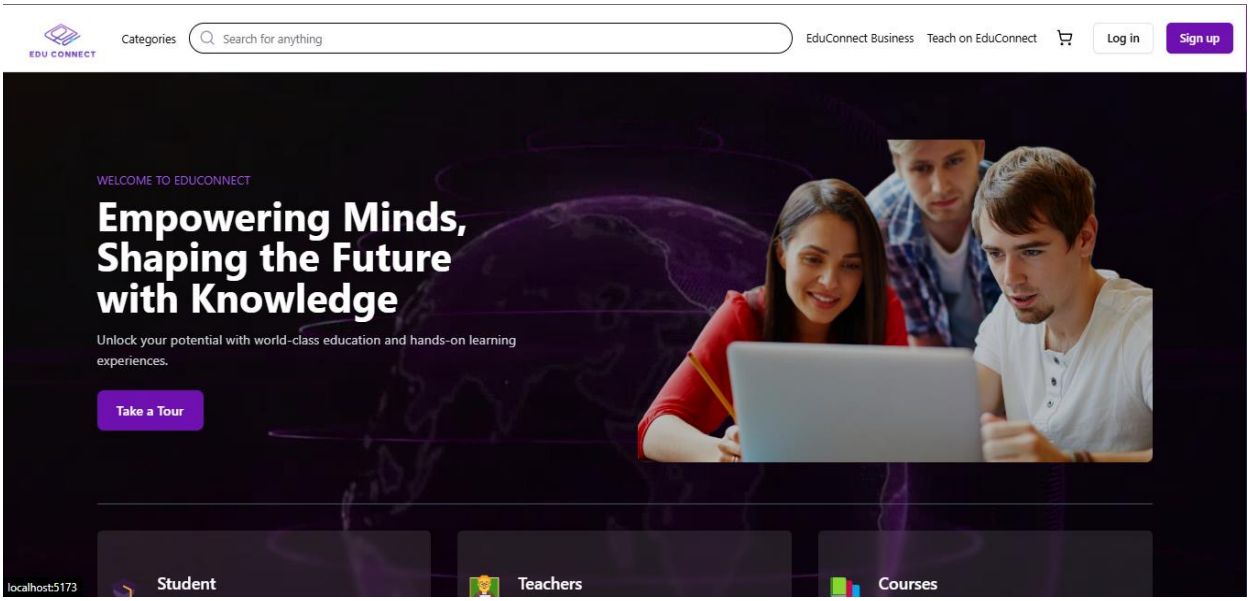
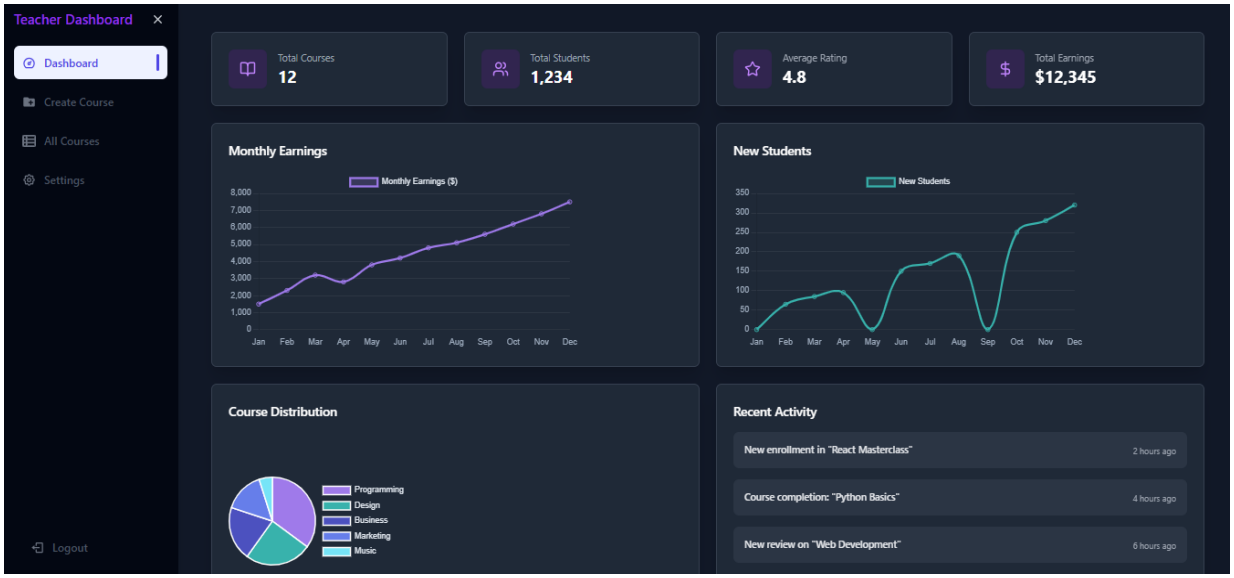


Figure 5.1 Home Page

This is the hero section of the homepage of our Project.

5.3 Teacher Dashboard:



This is Teacher dashboard UI. It shows the teacher's analytics.

Figure 5.2 About Page

This is the about page of our Project. It shows the developer names and its descriptions.

5.4 Students Portal

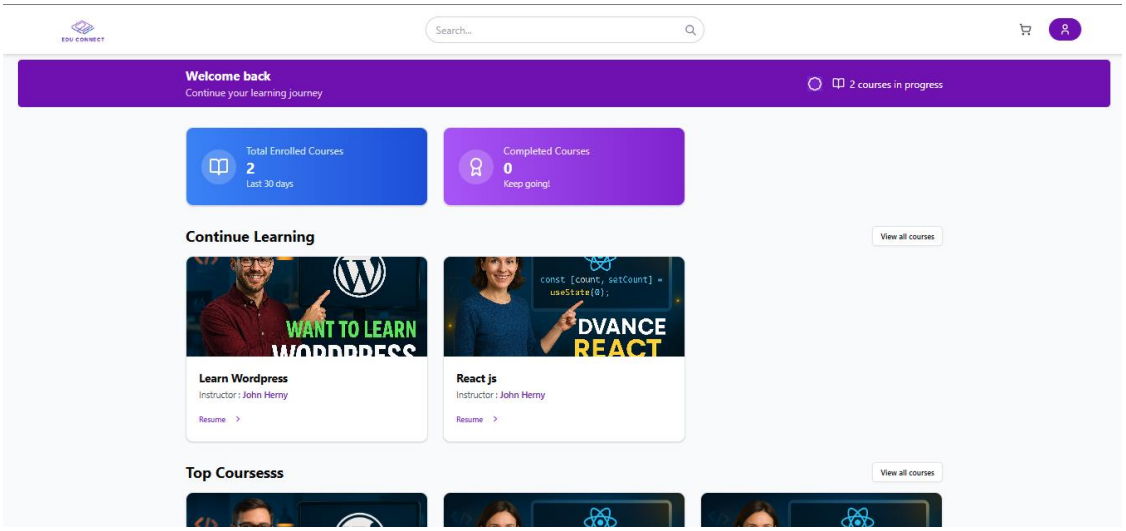


Figure 5.3 Student portal

This is our Student portal. The UI of student portal is user friendly easy to use and interactive. Where student can view their courses and they can start learning process.

5.5 Payment method

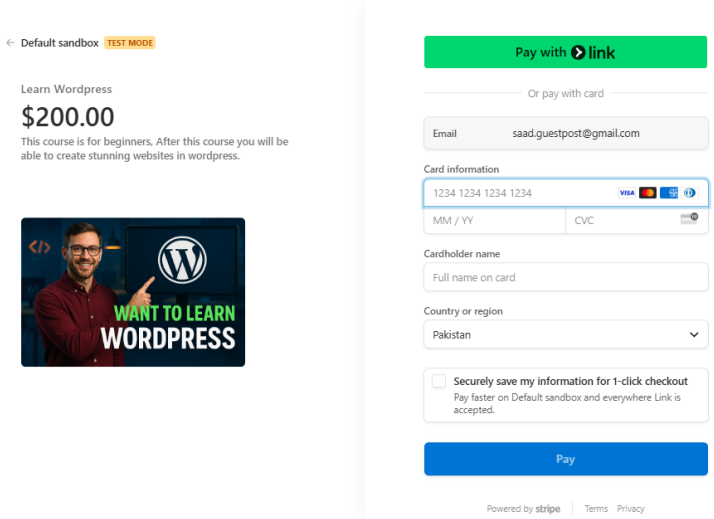


Figure 5.4 Payment method

In our Learning Management System (LMS), students can browse available courses and enroll in the ones they wish to purchase. To facilitate secure and efficient transactions, we have integrated **Stripe** as our primary payment gateway.

5.6 Discussion Board

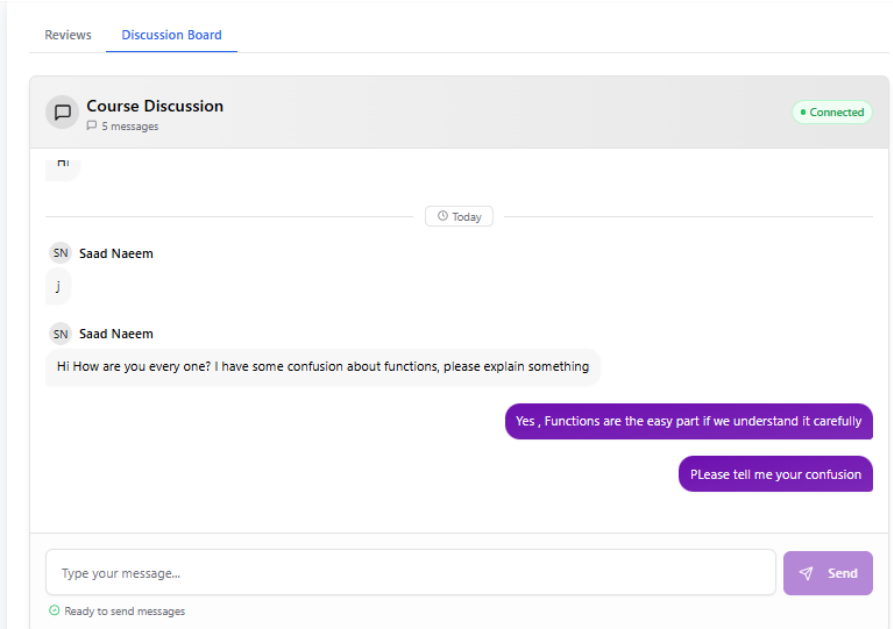


Figure 5.5 Discussion Board

5.7 Admin Dashboard

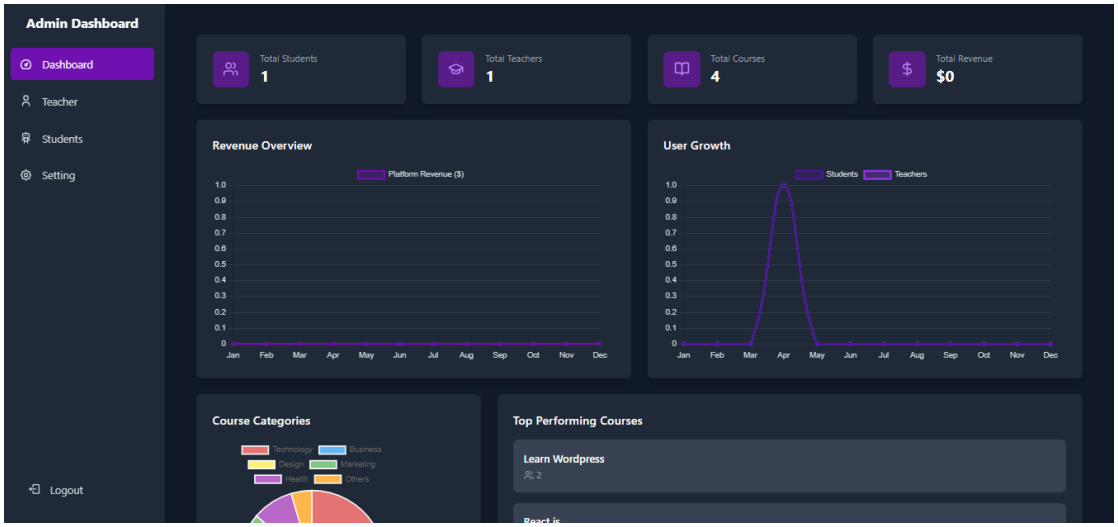


Figure 5.6 Admin dashboard

Admin dashboard here admin can check the course categories user growth and after all admin can check the revenue overview of the total courses.

CHAPTER 6
System Testing

6.1 Introduction

System testing is a crucial phase in the software development lifecycle that ensures the EduConnect web application operates as intended. It involves validating the complete integrated system, checking its functionality, performance, security, and usability under real-world conditions. The primary goal of system testing is to verify that all components of the application—frontend, backend, and database—work together seamlessly to deliver the expected user experience for both students and teachers :

Steps involve in system testing:

6.1.1 Data Verification and Preparation (Adapted for EduConnect):

- **EduConnect Context**:

The data in EduConnect refers to user information (students and teachers), courses, quizzes, discussion posts, and results.

- **Preprocessing**:

Verify that all data (course materials, user profiles, discussions, quiz results) are stored correctly in the database. Ensure data is properly structured (e.g., course ID, quiz ID, user roles) and preprocessed before being used in the system.

6.1.2 User role-based Testing:

EduConnect Context:

Divide the data into different testing sets, such as different user roles (students and teachers) and various courses or quizzes. This helps test the system across different user interactions and workflows, ensuring no data inconsistencies or errors.

6.1.3 System Integration Testing:

EduConnect Context: Verify that all components (frontend, backend, and database) are properly integrated. Ensure that features such as login, course management, quizzes, and discussion boards are linked together seamlessly. Check for proper communication between components.

6.1.5 Comprehensive System Testing:

- **Functional Testing**:
- Ensure that each functionality (e.g., user login, course enrollment, quiz attempts,

discussion board interactions) works as expected. Test role-based access control to ensure students and teachers have appropriate access to resources.

- **Non-Functional Testing**: Assess performance (loading times, responsiveness), security (data protection and encryption), and usability (ease of use across devices).
- **End-to-End Testing**: Simulate real-world workflows, such as a student enrolling in a course, attempting a quiz, and participating in a discussion, to ensure the system operates smoothly from start to finish.

6.1.5 Evaluating the System's Performance:

EduConnect Context: Evaluate the performance of the EduConnect web application under load conditions (e.g., multiple students taking quizzes at once). Monitor page load times, system stability, and the database's ability to handle simultaneous requests. Check if the system meets performance expectations under different conditions, including high traffic.

6.2 Test Scenario for EduConnect:

Test Scenario for Educonnect Web Application:

Data handling:

- **Test Case 1**: Verify that all course and user data are correctly uploaded into the system.
- **Test Case 2**: Ensure that all fields (e.g., user profiles, course materials) are correctly validated and formatted.

6.2.1 Authentication and Authorization:

- **Test Case 3**: Test the login system to verify that user credentials are properly authenticated and that session management works as expected.
- **Test Case 4**: Ensure that roles (student and teacher) have proper permissions for their respective functionalities (e.g., students can't create courses, teachers can).

6.2.3 User Interaction and Responsiveness:

- **Test Case 5**: Evaluate the system's responsiveness when students access courses, view materials, and submit assignments.
- **Test Case 6**: Test real-time features like discussions, ensuring smooth communication between users without delays.

6.2.4 Load and Scalability Testing:

- **Test Case 7**: Measure the system's scalability by simulating multiple concurrent users (students and teachers) accessing the platform simultaneously.

6.2.5 Deployment:

Test Case 8: After deployment, verify that the system integrates correctly with the production environment, including external tools or services (if applicable). Ensure that input data (e.g., course materials, quizzes) is processed correctly.

6.3 Test Plan

The purpose of this test plan is to ensure that the Educonnect Web Application of online learning platform is functioning correctly and producing accurate results.

6.3.1 Testing Objective:

The main objective of the test plan is to ensure the **accuracy** and **reliability** of the EduConnect system, focusing on:

- Proper user role access.
- Functionality across different modules (course management, quiz taking, etc.).
- Performance under real-world conditions.

6.3.4 Test Environment:

- **Hardware**: Testing will be conducted on various devices (desktop, tablet, mobile).
- **Software**: The system will be developed using **React.js**, **Node.js** with **Express**, and **MongoDB**. Testing tools include **Postman** for API testing.

6.3.5 Test Execution:

The tests will be executed in the following order:

1) Test Case 1
2) Test Case 2
3) Test Case 3
4) Test Case 4
5) Test Case 5
6) Test Case 6

6.4 Definition of Test Cases

6.4.1 Accuracy Testing:

Test Case 1:

Objective: To ensure that user data, course information, and quiz results are accurately stored and retrieved by the system.

Steps:

- Log in as a student and enroll in a course.
- Attempt a quiz and submit answers.
- Log in as a teacher and verify that the quiz results are correctly recorded and visible.
- Check if the course and quiz progress are correctly reflected in the user's profile.

- **Expected Result**:
    - The system should correctly display course enrollments, quiz results, and user profile information.

6.4.2 Performance Testing:

Test Case 2:

- **Objective**: To assess the performance of the application under normal and heavy loads.

Steps:

- Simulate multiple users accessing the platform concurrently (e.g., 50+ users accessing courses, quizzes, and discussion boards).
- Measure the load time of the website during peak usage.
- Measure the response time for quiz submissions and course updates.
- Track any system delays or crashes during high traffic.

Expected Result:

- The system should handle multiple concurrent users with minimal delay in response times.
- No crashes should occur under high load.

- Page load times should remain within acceptable limits.

Test Case 3:

6.4.3 Stability and Consistency Testing:

Objective: To ensure the system remains stable over time with repeated usage.

Steps:

- Log in as a student and start browsing courses, attempt quizzes, and post in discussions.
- Simulate the same actions over a prolonged period (e.g., multiple sessions over several hours or days).
- Monitor the system for any crashes, slowdowns, or unresponsiveness.
- Perform system restart and check for state consistency (user sessions, quiz progress, etc.).

Expected Result:

- The system should remain stable without crashes, slowdowns, or loss of data over extended use.
- User sessions and progress should be retained after system restarts

Test Case 4:

6.4.4 Usability Testing:

Objective: To ensure that the user interface is intuitive and the platform is easy to navigate for both students and teachers.

Steps:

- Log in as a student and navigate through the course catalog.
- Check if it's easy to enroll in a course, view course materials, and attempt a quiz.
- Log in as a teacher and verify the ease of creating and managing courses, assigning quizzes, and moderating discussion boards.
- Ensure the navigation is smooth and intuitive across different devices (desktop, tablet, and mobile).

Expected Result:

- Both students and teachers should be able to navigate the platform with ease, without requiring additional support.
- The layout should be responsive and work seamlessly on all device types.

Test Case 5:

6.4.5 Security Testing:

Objective: To verify that sensitive data (e.g., user credentials, grades, and personal information) is protected and that unauthorized users cannot access restricted resources.

Steps:

- Test the login system with valid and invalid credentials.
- Check for password hashing and secure storage of user data.
- Test access control by attempting to access teacher-only resources (such as course creation tools) while logged in as a student.
- Test data encryption for quiz results and personal information.

Expected Result:

- Unauthorized users should not be able to access restricted areas (e.g., a student should not be able to create or modify courses).
- User data should be securely stored and encrypted.
- The system should handle invalid login attempts appropriately and securely.

Test Case 6:

6.4.6 Integration Testing:

Objective: To ensure that the different modules of the application (frontend, backend, and database) work together as expected:

Steps:

- Log in as a student and check if course enrollment updates in real time across the frontend and backend.

- Test if quiz submissions correctly update in the database and show results on the student's dashboard.
- Verify that discussion board posts appear in real-time for both students and teachers.
- Check if user data (e.g., profile, quiz results) is consistently updated across the system after each action.

Expected Result:

- Data should flow smoothly between the frontend, backend, and database without any inconsistencies.

- All modules should be fully integrated and synchronize data in real time.

6.4.5 Test Reporting:

The test results will be reported in a standard format using a test report template. The report will include the following information:

- ✓ Test Summary: A summary of the test results, including the total number of tests executed, the number of tests passed, and the number of tests failed.
- ✓ Test Cases: A detailed description of each test case, including the test objective, the test steps, and the expected and actual results.
- ✓ Test Results: A summary of the test results, including the accuracy score, the model's performance under different conditions, and the stability and consistency of the model.
- ✓ Test Conclusions: A conclusion on the validity of the model, including any recommendations for further improvements or modifications.

Educonnect Web Application

CHAPTER 7
` Conclusion and future Work

7.1 Conclusion

EduConnect successfully bridges the gap between teachers and students by providing an all-in-one platform for course management, real-time interaction, quizzes, and certification. It offers a user-friendly interface built on the MERN stack, ensuring secure and efficient learning experiences. The system meets the core goals of online education by promoting accessibility, flexibility, and engagement.

7.2 Future Work

1. Dataset Expansion

To enhance personalization and recommendation systems, future iterations can include larger datasets on user behavior, course interactions, and learning outcomes.

2. Transfer Learning

By integrating transfer learning models, EduConnect can support advanced features like AI-driven student performance prediction and personalized learning paths.

3. Multi-Modal Sensing

Incorporating multi-modal inputs such as voice, handwriting (for assignments), or even facial expression recognition can make EduConnect more interactive and inclusive.

4. Real-Time Classification

Implementing real-time classification algorithms for detecting learning patterns or engagement levels during quizzes or lectures can improve adaptive learning.

5. Online Learning (AI/ML-Based)

Future upgrades may include AI-powered adaptive learning systems that adjust content difficulty based on student performance in real-time, enabling a personalized education experience.